

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

23rd November 2024

SP8 - General Insurance: Pricing

Time allowed: 3 Hours 15 Minutes

Total Marks: 100

Q. 1) The manager of the Mumbai Indians cricket team has asked your General Insurance Company for insurance for this season's Indian Premier League (IPL).

a) Suggest 8 products that can be offered to the team while describing their relevance. (8)

b) List the information required from the team manager for the purpose of pricing. (3)

The Chief Risk Officer of the Company is worried on whether your Company has the relevant expertise and data to price such a cover.

c) Explain how data will be arranged for the purpose of pricing this policy for Mumbai Indians. (2)

The reinsurer approached by the Company has denied any past experience of writing such business and has agreed to provide XoL cover of 5cr xs 2 cr with 1 free reinstatement and 2 paid reinstatements allowed.

d) Basis the cost information provided in the table below, suggest the total premium to be paid by direct insurer for complete reinsurance cover:

Limit	Cost in lakhs
0.5 Cr	1.00
0.5 Cr - 1 Cr	0.85
1 Cr - 2 Cr	1.20
2 Cr - 4 Cr	1.60
4 Cr - 5 Cr	0.50
5 Cr - 7 Cr	0.80
7 Cr - 10 Cr	1.00
10 Cr - 12 Cr	0.58
12 Cr - 17 Cr	1.00
17 Cr - 20 Cr	0.40
20 Cr - 22 Cr	0.13
22 Cr - 25 Cr	0.08

(3)
[16 Marks]

Q. 2) You are the Appointed Actuary of the second largest General Insurance Company, and your CEO is concerned about the company's declining market share. The initial analysis by the strategy team suggests that this decline is linked to competitors' increased focus on Crop Insurance this fiscal year. Although your company hasn't been active in Crop Insurance, the CEO is seeking your assistance to gain a better understanding of this area.

a) Define Measure of Exposure and comment on what should be the exposure measure for Crop Insurance and why? (3)

b) What are the considerations required to decide whether the Company can write Crop Insurance in future? (4)

c) What form of reinsurance would best suit this product for the Company and why? (6)

Your company has decided to enter the Crop Insurance market next year and is currently seeking XoL (Excess of Loss) reinsurance support. A junior team member is analyzing published industry data on Crop Insurance claims using the Frequency-Severity Method but is uncertain about the credibility of the available data.

- d) Explain why number of claims for determining credibility in Crop Insurance is expected to be lower than that in Motor Own Damage claims. (4)
 - e) Explain why the above method of credibility will only be applicable to attritional losses and not to Catastrophic claims and suggest an approach required for catastrophic claims. (4)
- [21 Marks]**

Q. 3) You are the Chief Actuary of a large & traditional General Insurance Company. Recently a new competitor has entered the market branding themselves as a “Digital Only Company” and airing numerous television advertisements claiming to offer INR 20,000 lower premium for Motor Insurance policies. Your Chief Distribution Officer has requested a detailed note on the following.

- a) Why can't your Company not run television advertisements?
- b) How can the premium difference be so high?

Write contents of the note for clarification.

[8 Marks]

Q. 4) Your neighbor's son, who is studying statistics and aims to advance his career through Actuarial Science, is conducting research. As part of his study, he inquires about your daily responsibilities at work.

- a) As a Chief Pricing Actuary of a leading General Insurance Company, explain the spectrum of Actuarial analyses you are responsible for at work. (10)
- b) Given all the probability work you do, he's curious if your expertise could help in winning a casino lottery. Using detailed workings of stochastic aggregate distribution, explain how analyzing claims differs from wagering. (4)

[14 Marks]

Q. 5) The Regulator has recently removed tariff regulations from Motor Third Party pricing within the General Insurance Industry. As the Pricing Actuary of a large multi-line insurer, you are tasked with pricing the product using historical internal data.

For the initial year, the Regulator has mandated the Burn Cost approach for pricing. All Premium & Claim Amounts given below are in Crores.

- a) Your Claims Head has enquired how many years of past claims data would you use and why? (2)
- b) Your team has asked whether the data should be extracted on Accident Year level or Reporting Year level. Explain with reasons which would be selected and why? (2)

Below is a snapshot of internal data prepared by the claims team:

AY	Premium (adj. for price increase)	Paid Claims	Statistical Reserve	Case Reserve
2016-17	1,649	825	650	577
2017-18	1,929	1,447	450	386
2018-19	2,238	694	600	560
2019-20	2,350	635	750	705
2020-21	3,032	940	1,100	1,061
2021-22	3,759	827	1,550	1,504
2022-23	4,210	800	1,950	1,895
2023-24	5,347	535	2,700	2,673

- c) The Claims Head wants to understand why have you prepared the data adjusted for premium increase? (1)
- d) Your team is curious to know why the Company holds 2 types of outstanding reserve and which type should be used for pricing further. (3)

Additionally, below are the inputs received from the Reserving Team:

AY	Premium	IBNR Factor
2016-17	1,649	13.0%
2017-18	1,929	14.5%
2018-19	2,238	17.5%
2019-20	2,350	22.0%
2020-21	3,032	27.0%
2021-22	3,759	29.0%
2022-23	4,210	33.0%
2023-24	5,347	35.0%

- e) Considering the earliest year of available claims data also has some IBNR factor, the team wants to know whether this claims data can be used for the purpose of pricing. (1)

On analysing the emerging trends in incurred loss ratio, the team found a considerable reduction since 2021-22. On enquiring, the Chief Underwriting Officer (CUO) has confirmed that this is due to reduction of Commercial Vehicle (CV) business written during that period.

Your CUO has provided this additional information to this regard.

AY	CV Premium %	CV ULR%
2016-17	20%	110%
2017-18	20%	110%
2018-19	20%	110%
2019-20	20%	110%
2020-21	20%	110%
2021-22	10%	110%
2022-23	10%	110%
2023-24	10%	110%

- f) Your team wants your guidance on the adjustments to be made to the available internal data,

to increase its relevance for future time periods.

(4)

- g) Basis the adjustments mentioned above, calculate the appropriate Ultimate Claim Amount for the purpose of pricing Motor Third Party product going forward (ignore claims inflation)
(12)

Below is the additional information available from Sales Team:

AY	Exposure
2016-17	9,349
2017-18	10,938
2018-19	11,228
2019-20	11,790
2020-21	15,209
2021-22	17,961
2022-23	20,116
2023-24	25,547

- h) If your Company is expecting the next year to be an average of previous 8 years, what should be the price increase for the next year to achieve 65% Loss Ratio? (4)
- i) When you propose this increase in premium to your Sales Head, he expresses that this is too big a change to be absorbed in a single year and asks if we can spread it as a 2% incremental impact for next 3years, everything else being equal. If required, suggest alternate rationale to your Sales Head. Assume that experience in next 2 years will be average of all past years. (Ignore time value of money)

AY	Exposure
2024-25	26,824
2025-26	28,166
2026-27	29,574

(12)
[41 Marks]
